



How Much Do Fans Love Taylor Swift? Day 1



Taylor Swift wants to find out how much fans enjoyed the Eras Tour shows. She will ask fans, "From 1 to 100, where 100 is the most, how much did you enjoy the concert?" The V.I.P. section she wants to survey has 50 seats (5 rows x 10 columns). The stage runs along the northern edge of the venue (where Taylor is pictured). She wants to take a sample of 10 fans.

1. **Method #1:**

Take a simple random sample (SRS) of 10 fans. Explain the steps.



2. **Method #2:**

Randomly choose 2 fans from each horizontal row.



3. **Method #3:**

Randomly choose 1 fan from each vertical column.



4. Which method do you think will give the best estimate for the average enjoyment? Why?



Now, it's time for the actual data. For each of your samples on the previous page, calculate the average enjoyment. Add your average to the dotplots on the board.



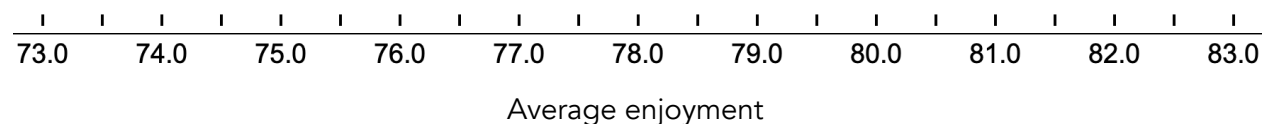
Sample #1:

92	89	90	88	95	100	98	93	95	84
82	86	90	88	86	91	90	89	85	83
80	74	80	67	81	82	76	77	74	65
72	68	74	73	70	69	72	70	68	67
69	67	68	68	64	66	63	63	70	68

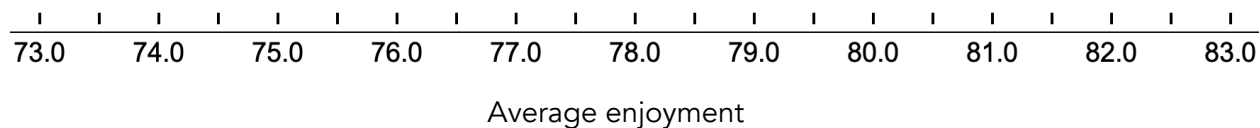
Sample #2:

Sample #3:

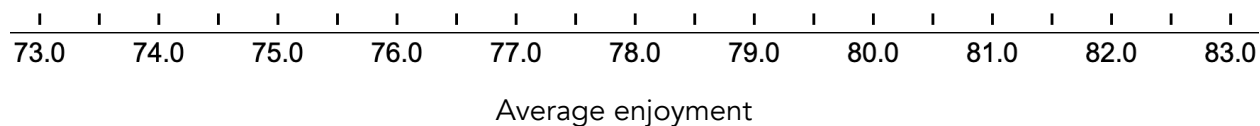
Method #1: Simple Random Sample (SRS)



Method #2: Stratify by Row



Method #3: Stratify by Column



5. Based on the dotplots above, which method will produce the best estimate for the true population average enjoyment? Explain.

Lesson 2.2 – Stratified Random Samples

QuickNotes

Check Your Understanding

A principal at a large high school is proposing a new digital hall pass that students would access using their phones. She wants to know what proportion of the student body is in favor of the proposed change. She will take a sample of 100 students.

- a. Would this sample be with replacement or without replacement? Explain.

- b. Explain how the principal could take a stratified random sample, using grade level as strata.

- c. Explain why the stratified random sample might produce a better estimate than a simple random sample.